









Statistics of Financial Markets

Chapter 15: Systemic risk



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Acronyms used in this material (1/2)

- **AI**: Artificial Intelligence
- **AIG**: American International Group
- **ASF**: Autoritatea de Supraveghere Financiară (the Financial Supervisory Authority of Romania)
- **AT1**: Additional Tier 1 (capital instruments)
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **BIS**: Bank for International Settlements
- **BNP**: Banque Nationale de Paris (BNP Paribas) — the French banking group BNP Paribas
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- **CET1**: Common Equity Tier 1 (capital)
- **CNSM**: Comitetul Național pentru Supravegherea Macroprudențială (the National Committee for Macroprudential Oversight)
- **COVID-19**: COronaVIrus Disease 2019
- **DGC**: Degree of Granger Causality
- **ECB**: European Central Bank
- **EODHD**: EOD Historical Data (market-data provider)
- **ES**: Expected Shortfall
- **ESRB**: European Systemic Risk Board

Acronyms used in this material (2/2)

- **EU**: European Union
- **FINMA**: Eidgenössische Finanzmarktaufsicht (the Swiss Financial Market Supervisory Authority)
- **FRM**: Financial Risk Meter
- **FSB**: Financial Stability Board
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **HSBC**: Hongkong and Shanghai Banking Corporation
- **IFRS**: International Financial Reporting Standards
- **IMF**: International Monetary Fund
- **ING**: Internationale Nederlanden Groep (the Dutch banking group ING)
- **JPM**: JPMorgan Chase (ticker symbol)
- **LRMES**: Long-Run Marginal Expected Shortfall
- **MES**: Marginal Expected Shortfall
- **NYU**: New York University
- **OLS**: Ordinary Least Squares
- **SRISK**: Systemic RISK (capital shortfall in a crisis)
- **SVB**: Silicon Valley Bank
- **TENET**: Tail-Event driven NETwork risk
- **UBS**: Union Bank of Switzerland (former name of UBS)
- **US**: United States
- **VAR**: Vector AutoRegression
- **VIX**: CBOE Volatility IndeX

Today's Question and Route

- **Question**: when does the loss of one bank become a loss of the whole financial system, and how do we measure it from market data?
  - Chapter 10 measured the risk of one position (VaR 1%, ES 2.5%); this chapter measures the risk that one institution adds to the system
- **Route** of the chapter
  - what systemic risk is: contagion, common exposures, fire sales, too big and too interconnected to fail
  - four crises in the data: 2008, the euro area in 2011–2012, March 2020, March 2023
  - measures: correlation and tail dependence, CoVaR and ΔCoVaR, MES and SRISK, networks
  - macroprudential policy: Basel III buffers, the ESRB, the BNR

Learning Outcomes

- Explain the channels of systemic risk and recognise them in the crises of 2008, 2011–2012, 2020 and 2023
- Estimate CoVaR 1% and ΔCoVaR by linear quantile regression and interpret them next to VaR 1%
- Compute MES 5%, LRMES and SRISK, and explain the role of leverage
- Distinguish contagion from interdependence (the Forbes–Rigobon correction) and read a tail-dependence estimate
- Read a Granger-causality network and a Diebold–Yilmaz connectedness table
- Name the macroprudential tools of Basel III and the institutions that apply them in the EU and in Romania

Reading and Tools

- Textbook: Franke, Härdle & Hafner (2019), Ch. 16 (VaR) and Ch. 17 (copulae); surveys: Bisias, Flood, Lo & Valavanis (2012); Benoit, Colliard, Hurlin & Pérignon (2017)
  - the landmark papers of the chapter: Adrian & Brunnermeier (2016); Acharya, Pedersen, Philippon & Richardson (2017); Brownlees & Engle (2017); Billio, Getmansky, Lo & Pelizzon (2012); Diebold & Yilmaz (2014)
- Python Quantlets of this chapter: Quantlets/Ch_15
- Lecture notebook: open in Google Colab
- Video courses: Financial Risk Meter for Emerging Markets; Dynamic Crypto Networks

A Definition

- **Systemic risk**: the risk that the financial system as a whole stops working (lending, payments, trading), with large costs for the real economy IMF, BIS & FSB (2009)
  - the definition of the IMF (International Monetary Fund), the BIS and the FSB (Financial Stability Board) for the G-20 in 2009
- Three criteria for a **systemically important** institution IMF, BIS & FSB (2009)
  - **size**: how much of the system's services it provides
  - **substitutability**: whether others can take over its activity quickly
  - **interconnectedness**: how many others it would drag down
- Key point: systemic risk is a property of the **system**, not of a single balance sheet

Why VaR of Each Bank Is Not Enough

- VaR 1% of a bank (Chapter 10): the loss of **this** bank exceeded with probability 1%
  - it ignores what happens to the others when this bank is in trouble
- **Fallacy of composition**: if every bank sells risky assets to cut its own VaR, prices fall for all and the risk of all rises
  - a small bank with a high VaR can matter less than a large, connected bank with a moderate VaR
- We need measures of the **tail of the system given the institution**, or of the **institution given the tail of the system**
  - CoVaR: the system given that the bank is in distress Adrian & Brunnermeier (2016)
  - MES and SRISK: the bank given that the market crashes Acharya, Pedersen, Philippon & Richardson (2017); Brownlees & Engle (2017)

Channel 1: Direct Contagion

- **Contagion**: the failure of one institution causes losses or failures at others
  - direct exposures: interbank loans, derivatives, deposits held at other banks
  - a network of balance sheets: a loss travels along the links Allen & Gale (2000)
- Network structure matters Allen & Gale (2000); Haldane & May (2011)
  - many links spread small shocks thin, so the system absorbs them
  - the same links transmit large shocks everywhere: "robust yet fragile"
- **Too interconnected to fail**: an institution whose failure would hit many counterparties at once

Channel 2: Common Exposures and Fire Sales

- **Common exposures**: banks hold the same assets (mortgages, government bonds), so one shock hits all of them at once
  - no link between the banks is needed: the correlation comes from the assets
- **Fire sales**: a bank that must sell quickly sells below the fundamental value Shleifer & Vishny (2011)
  - the lower price is a loss for every holder of the asset (marked to market)
  - **liquidity spiral**: losses $\rightarrow$ higher margins $\rightarrow$ more sales $\rightarrow$ lower prices Brunnermeier & Pedersen (2009)
- **Leverage** amplifies everything: with assets 20 times equity, a 5% fall in asset value wipes out the equity

Channel 3: Bank Runs and Information Contagion

- **Bank run**: depositors withdraw at once because they fear that others will withdraw first Diamond & Dybvig (1983)
  - a bank holds long, illiquid loans and short, liquid deposits: it cannot pay everyone at once
  - a run can hit a solvent bank; deposit insurance removes the reason to run
- **Information contagion**: trouble at one bank makes investors doubt similar banks
  - Northern Rock (United Kingdom), September 2007: the first run on a British bank in over a century



Depositors queue outside a Northern Rock branch, September 2007

Photo: Dominic Alves (2007); CC BY 2.0; Wikimedia Commons

Too Big to Fail

- **Too big to fail**: a bank whose failure would damage the economy so much that the state rescues it
  - the expected rescue lowers its funding cost: an implicit subsidy
  - moral hazard: the bank takes more risk because losses are shared with taxpayers
- The regulatory answer after 2008 BCBS (2018)
  - G-SIBs (global systemically important banks): a list published every year, with extra capital for each bank
  - resolution plans: a failing bank must be wound down without taxpayers' money
- Market measures (CoVaR, SRISK) let us rank banks without waiting for a failure

Measures of This Chapter at a Glance

Measure	Question	Tool
correlation, tail dependence	do banks and markets crash together?	pseudo-observations, t copula
CoVaR, ΔCoVaR	how much worse is the system when the bank is in distress?	quantile regression
MES	how much does the bank lose on the market's worst days?	conditional average
SRISK	how much capital would the bank miss in a crisis?	MES, leverage
networks	who transmits shocks to whom?	Granger tests, variance decompositions

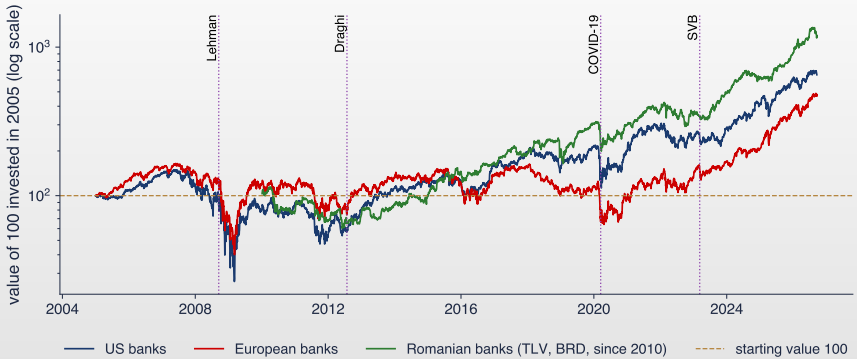
- All measures use only market prices: they are available daily, for any listed bank
- Their weakness: market prices can be calm while risk builds up (2006–2007)

Data of the Chapter

Group	Banks	Market index
US banks	JPMorgan Chase, Bank of America, Citigroup, Goldman Sachs, Morgan Stanley, Wells Fargo	S&P 500
European banks	Deutsche Bank, BNP Paribas, Santander, ING, HSBC	Euro Stoxx 50
Romanian banks	Banca Transilvania (TLV), BRD (since 2010)	BET

- Daily adjusted closes of the banks and closes of the indices from EODHD, January 2005 – 18 September 2026; the VIX as a stress indicator
- Prices are joined on common days first, then daily log returns in %: $r_t = 100(\ln P_t - \ln P_{t-1})$
- BVB banks from 2010: before, many days have no trades; TLV without 30–31 May 2016 (a price adjustment applied one day late, Chapter 2)
- Bank portfolio of a group: equal weights, rebalanced daily

Bank Portfolios since 2005



- Value of 100 invested in each equal-weighted portfolio (log scale); dotted lines: Lehman, Draghi's speech, COVID-19, the failure of SVB
- Maximum drawdown: US banks −82% (6 March 2009), European banks −76% (9 March 2009), Romanian banks −47% (5 June 2012)

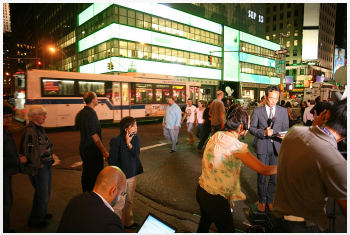
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Interpretation of the Bank Portfolios

- 2008–2009: US and European banks lost about four fifths of their value; the trough came in March 2009, six months after Lehman
  - a systemic crisis is a process that lasts months, not a single bad day
- European banks stayed below their 2005 level for most of the period: the euro-area crisis and low interest rates
- Romanian banks: their worst fall came in the euro-area crisis of 2011–2012 (the parent banks of many Romanian banks are in the euro area)
- March 2020 and March 2023: sharp falls, but short; central banks and governments reacted within days

2008: Lehman Brothers

- 15 September 2008: Lehman Brothers files for bankruptcy, the largest bankruptcy in US history (assets above 600 billion USD)
  - 16 September: AIG, an insurer that had sold protection on mortgage debt to many banks, is rescued by the Federal Reserve
  - the same day, a money market fund holding Lehman debt "breaks the buck" and investors run from such funds
- All three channels at once
  - direct exposures (derivatives with Lehman), common exposures (mortgages), a run on short-term funding
- From the end of August to the trough in November: US banks −59%, European banks −54%, S&P 500 −41%



Lehman Brothers headquarters, New York, 15 September 2008

Photo: Robert Scoble (2008); CC BY 2.0; Wikimedia Commons

2010–2012: The Euro-Area Crisis

- Greece (2010), then Ireland, Portugal, Spain and Italy: doubts about government debt
  - **the bank–sovereign loop**: banks hold their own government's bonds, and the government stands behind its banks
  - a fall in bond prices weakens the banks; rescuing the banks weakens the state
- 26 July 2012: Mario Draghi, President of the ECB (European Central Bank): "the ECB is ready to do whatever it takes to preserve the euro" Draghi (2012)
  - a promise, not a purchase, stopped the spiral: the role of expectations
- July 2011 – September 2012, worst point: European banks −41%, US banks −37%, Romanian banks −34%

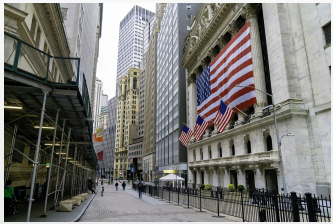


The ECB building in Frankfurt am Main

Photo: DXR (2015); CC BY-SA 4.0; Wikimedia Commons

March 2020: The Dash for Cash

- COVID-19 lockdowns: everyone wanted cash at the same time, even government bonds were sold
  - the stress started outside the banks: funds, firms, the bond market
  - banks were better capitalised than in 2008 (Basel III) and became part of the solution (credit lines)
- Fast policy response: central banks bought assets and lent in dollars to other central banks
  - trough on 23 March 2020: US banks −48%, European banks −48%, Romanian banks −35%, S&P 500 −34%



The Financial District of New York, 25 March 2020

Photo: Billie Grace Ward (2020); CC BY 2.0; Wikimedia Commons

March 2023: SVB

- SVB (Silicon Valley Bank): deposits of technology firms, mostly above the insured limit, invested in long-term bonds
  - rising interest rates in 2022 created large unrealised losses on the bonds
  - 9 March 2023: depositors tried to withdraw about 42 billion USD in one day; 10 March: the bank is closed Federal Reserve (2023); FDIC (2023)
- A run in the age of mobile banking: faster than any run before
  - a mid-sized bank became systemic through information contagion: other regional banks fell the next days



Silicon Valley Bank headquarters, Santa Clara, 13 March 2023

Photo: Minh Nguyen (2023); CC BY-SA 4.0; Wikimedia Commons

March 2023: Credit Suisse

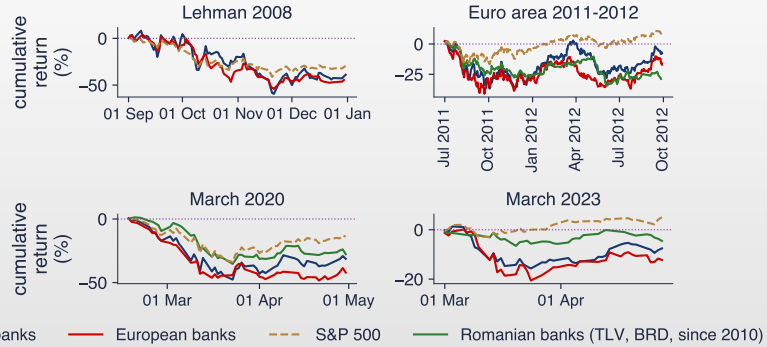
- Credit Suisse: a G-SIB that had lost clients and money for years (scandals, losses on hedge-fund clients)
  - after SVB, investors and depositors left quickly
- 19 March 2023: takeover by UBS, approved by the Swiss supervisor FINMA FINMA (2023)
  - AT1 bonds (additional tier 1 capital) of about 16 billion CHF were written down to zero
  - a rescue by merger, not a resolution: too big to fail was still alive
- From the end of February to the trough: European banks −21%, US banks −16%, Romanian banks −7%



The Credit Suisse headquarters at Paradeplatz, Zürich

Photo: Thomas Wolf, www.foto-tw.de (2013); CC BY-SA 3.0 DE; Wikimedia Commons

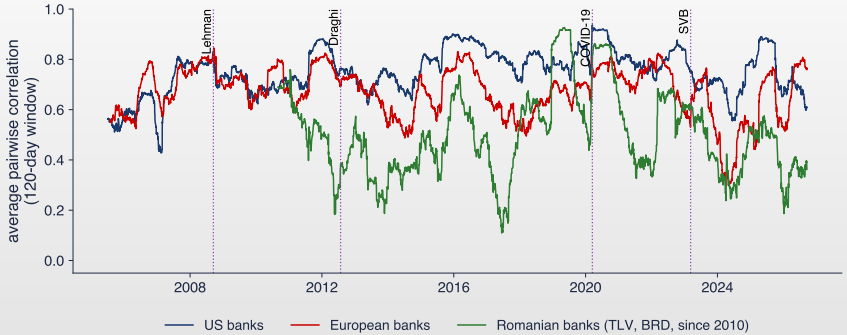
Four Episodes Side by Side



- Cumulative return from the start of each window; the Romanian banks enter the data in 2010
- Interpretation: in 2008 and 2011–2012 the banks fell much more than the S&P 500 (the banks were the problem); in 2020 they fell with the market; in 2023 they fell while the market rose

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Correlation between Banks Rises in Crises



- Average of the pairwise correlations of daily returns within each group, on 120-day windows
- US banks: 0.59 before the crisis (July 2005 – June 2007), up to 0.94 (16 March 2020); TLV and BRD: 0.31 in 2017, 0.85 in April–June 2020, maximum 0.93 (9 May 2019)

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Interpretation: When Diversification Fails

- In crises, the banks move together: a portfolio of several banks is almost one bank
  - the common factor (the market, funding stress) dominates the bank-specific news
- The two Romanian banks: low correlation in 2017, very high in 2020; the maximum (9 May 2019) falls outside any crisis
  - in calm years the BVB stocks are driven by their own news and by low liquidity; in a crisis, by the common shock
- Caution: a rolling correlation estimated on 120 days is noisy (for two stocks, it can peak in a calm year), and a rise in correlation does not prove contagion (next slide)

Contagion or Interdependence? Forbes and Rigobon (1/2)

- If $y = \beta x + e$ with fixed β, the correlation of x and y rises **automatically** when the variance of x rises Forbes & Rigobon (2002)
  - x: the return of the source market; y: the return of the other market; β: the strength of the link; e: a shock of y unrelated to x
  - a higher correlation in a crisis may be the same link seen in a more volatile period: **interdependence**
  - **contagion**: the link itself becomes stronger
- Corrected correlation:

$$\rho^* = \frac{\rho}{\sqrt{1 + \delta(1 - \rho^2)}}, \qquad \delta = \frac{\sigma^2_{\text{crisis}}}{\sigma^2_{\text{calm}}} - 1$$

- Notation
  - ρ: the correlation measured in the crisis; σ^2: the variance of the source market in each period
  - δ: the relative rise of that variance; ρ^*: the crisis correlation that the calm-period volatility would give
  - compare ρ^* with ρ_{calm}: $\rho^* > \rho_{\text{calm}}$ points to contagion

Contagion or Interdependence? Forbes and Rigobon (2/2)

- Worked example
  - US bank portfolio (source) and European bank portfolio: calm January 2005 – June 2007 ($n = 506$), crisis 15 September – 31 December 2008 ($n = 72$)
  - $\rho_{\text{calm}} = 0.40$, $\rho_{\text{crisis}} = 0.60$; variance $0.89 \to 75.4$, so $\delta = 83.7$
  - $\rho^* = 0.60/\sqrt{1 + 83.7 \times 0.638} = 0.60/\sqrt{54.4} = 0.08$
- Interpretation: after the correction there is no rise in correlation: the data are consistent with interdependence, not with a stronger link

Tail Dependence (Chapter 10) (1/2)

- **Pseudo-observations**: $U = \text{rank}/(n+1)$, the rank of a return among the n returns of its series, scaled to $(0, 1)$
  - $U \leq 0.05$: one of the worst 5% of days of that series; the marginal distributions no longer matter
- **Lower tail dependence** at level q: $\lambda(q) = P(U_1 \leq q \mid U_2 \leq q)$
  - U_1: the bank; U_2: the market; $\lambda(q)$: the probability that the bank is in its worst q days, given that the market is in its worst q days
  - under independence $\lambda(q) = q$; the limit $\lambda_L = \lim_{q \to 0} \lambda(q)$ is the tail-dependence coefficient
- For systemic risk: tail dependence measures how often a bank and the market are in their worst days **together**

Tail Dependence (Chapter 10) (2/2)

- The t copula (Chapter 10) has a positive tail-dependence coefficient Demarta & McNeil (2005); the Gaussian copula has $\lambda_L = 0$

$$\lambda_L = 2\, t_{\nu+1}\Big(- \sqrt{(\nu+1)(1-\rho)/(1+\rho)}\Big) > 0$$

- Notation
  - $t_{\nu+1}$: the distribution function of a Student-t with $\nu + 1$ degrees of freedom
  - ρ: the correlation parameter of the copula; ν: its degrees of freedom; a small ν gives strong joint crashes
- Estimation
  - $\rho = \sin(\pi\tau/2)$ from Kendall's τ, a rank correlation; ν by maximum likelihood
  - $\tau = (C - D)/\binom{n}{2}$; C, D: the numbers of concordant and discordant pairs of days (both series move in the same or in opposite directions)

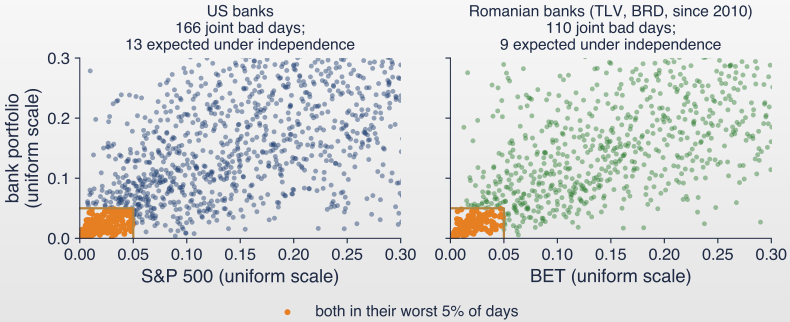
Tail Dependence of Banks and Markets

Pair	n	corr.	$\lambda(5\%)$	$\lambda(1\%)$	$\hat{\nu}$	λ_L (t)
US banks, S&P 500	5 229	0.79	0.64	0.63	3.2	0.49
European banks, Euro Stoxx 50	5 212	0.86	0.72	0.67	3.1	0.59
Romanian banks, BET	3 414	0.86	0.65	0.62	4.3	0.53
European banks, US banks	4 895	0.59	0.43	0.50	3.0	0.36

- $\lambda(q)$: empirical; $\hat{\nu}$ and λ_L: t copula; under independence $\lambda(5\%) = 0.05$ and $\lambda(1\%) = 0.01$
- Interpretation: on more than half of the market's worst days the banks are also in their worst days; the small $\hat{\nu}$ (about 3–4) means strong joint crashes, which a Gaussian copula misses entirely
- $\lambda(1\%)$ rests on a few dozen days: an imprecise estimate

SFM_ch15_dependence

Joint Bad Days



- Lower-left corner of the pseudo-observations; orange: both the bank portfolio and the index in their worst 5% of days
- US: 166 joint bad days against 13 under independence (about 13 times more); Romania: 110 against 9

SFM_ch15_dependence

From VaR to CoVaR

- X^i: return of institution i; X^{sys}: return of the system; level α = probability of the tail (Chapter 10)
  - $q_\alpha(X^i)$: the α-quantile of X^i; $\mathrm{VaR}^i_\alpha = -q_\alpha(X^i)$: the loss of i exceeded with probability α
- **CoVaR** (conditional VaR of the system) Adrian & Brunnermeier (2016)

$$\mathrm{CoVaR}^{sys|i}_\alpha = -q_\alpha\left(X^{sys} \mid X^i = q_\alpha(X^i)\right)$$

  - $q_\alpha(X^{sys} \mid \cdot)$: the α-quantile of the system return, conditional on what follows the bar
  - the VaR 1% of the system on a day when institution i is at its own VaR 1%: "CoVaR 1%"
- **ΔCoVaR**: the contribution of i to systemic risk

$$\Delta\mathrm{CoVaR}^{sys|i}_\alpha = \mathrm{CoVaR}^{sys|X^i=q_\alpha}_\alpha - \mathrm{CoVaR}^{sys|X^i=q_{50\%}}_\alpha$$

  - $q_{50\%}$: the median of X^i, a normal day of the institution
  - reading: how much the VaR of the system rises when i moves from a normal day (its median) to distress

Quantile Regression (Chapter 13)

- Linear quantile regression Koenker & Bassett (1978): $q_\alpha(Y \mid X = x) = a_\alpha + b_\alpha x$
  - a_α: the intercept (constant term); b_α: the slope
  - estimated by minimising $\sum_t \rho_\alpha(y_t - a - bx_t)$, with the check loss $\rho_\alpha(u) = u(\alpha - 1\{u < 0\})$
  - u: the residual; $1\{u < 0\}$: 1 if the point is below the line, 0 otherwise
  - for $\alpha = 1\%$, a residual below the line costs 99 times more than one above it: the line passes below 99% of the points
- Same idea as VaR 1% by quantile regression in Chapter 13, now with another institution as the regressor
  - the slope b_α can differ from the OLS slope: the tail can react more strongly than the centre

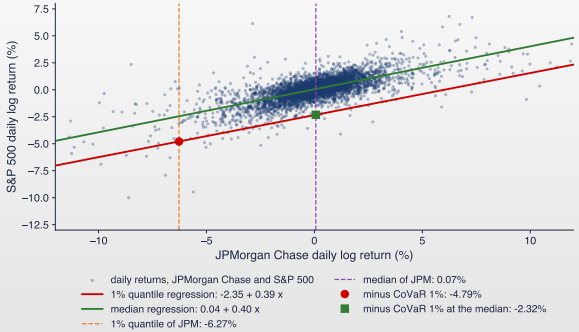
Estimating CoVaR in Three Steps

- **Step 1**: quantile regression of the system on the institution at level α: $X_t^{sys} = a + bX_t^i + \varepsilon_t$ Adrian & Brunnermeier (2016)
  - $\hat{a}$, $\hat{b}$: the estimated intercept and slope; the α-quantile of the error ε_t is zero by construction
- **Step 2**: the empirical quantiles of the institution: $\hat{q}_\alpha(X^i)$ (distress) and $\hat{q}_{50\%}(X^i)$ (median)
- **Step 3**: plug in
  - $\mathrm{CoVaR}_\alpha = -(\hat{a} + \hat{b}\,\hat{q}_\alpha(X^i))$
  - $\Delta\mathrm{CoVaR}_\alpha = \hat{b}\left(\hat{q}_{50\%}(X^i) - \hat{q}_\alpha(X^i)\right)$: only the slope and the distance between the two quantiles matter
- Here the system is proxied by the regional equity index (S&P 500, Euro Stoxx 50, BET); Adrian and Brunnermeier use the portfolio of all financial institutions

Worked Example: JPMorgan Chase and the S&P 500

- Daily log returns, 2005–2026 ($n = 5\,229$); quantile regression at 1%: $\hat{a} = -2.35$, $\hat{b} = 0.39$
  - JPM: $\hat{q}_{1\%} = -6.27\%$ (so VaR 1% of JPM $= 6.27\%$), median $\hat{q}_{50\%} = 0.07\%$
- **CoVaR 1%** $= -(-2.35 + 0.39 \times (-6.27)) = 2.35 + 2.445 = 4.795\%$
  - from the unrounded estimates: 4.79% (the value in the chart); CoVaR at the median of JPM: 2.32%
- **ΔCoVaR 1%** $= 0.39 \times (0.07 - (-6.27)) = 0.39 \times 6.34 = 2.47\%$
  - the unconditional VaR 1% of the S&P 500 is 3.53%
- Interpretation: when JPM is in distress, the loss of the index exceeded with probability 1% is 4.79%, against 2.32% on a normal JPM day

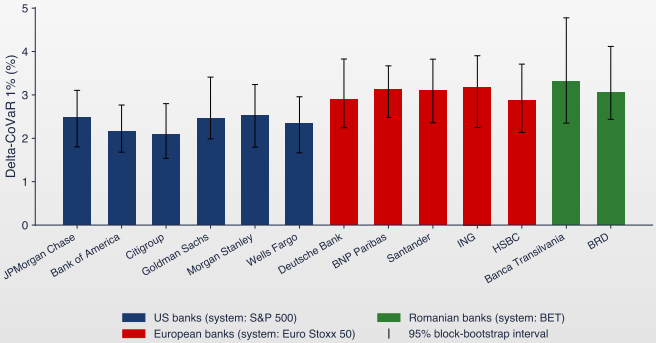
CoVaR in a Picture



- Red line: the 1% quantile of the S&P 500 given JPM; green line: the median; dashed: the 1% quantile and the median of JPM
- ΔCoVaR is the vertical distance between the red dot and the green square on the red line: 2.47%

SFM_ch15_covar

ΔCoVaR 1% of 13 Banks



- System: the regional index; 95% intervals from a moving-block bootstrap (blocks of 20 days, 200 resamples)
- US banks: 2.1–2.5%; European banks: 2.9–3.2%; TLV 3.31%, BRD 3.05%; largest: Banca Transilvania

SFM_ch15_covar

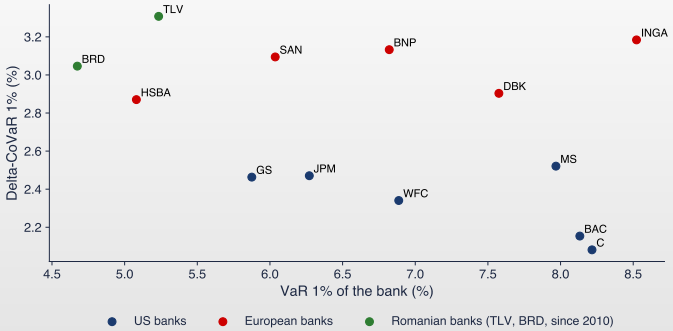
Interpretation of ΔCoVaR

- The intervals overlap: within a region, the data cannot rank the banks precisely
  - a ranking of systemic importance needs its uncertainty, like any estimate (bootstrap, Chapter 2)
- European and Romanian banks have a larger ΔCoVaR than US banks
  - banks weigh more in the Euro Stoxx 50 and in the BET than in the S&P 500: the index is closer to a banking system
  - TLV and BRD are among the largest stocks of the BET, so the BET partly measures themselves
- Robustness: the ranking at 5% is close to the ranking at 1% (Spearman correlation 0.87)
  - Spearman: the correlation of the ranks, $\rho_S = 1 - 6\sum_i d_i^2/(N(N^2-1))$; d_i: the difference of the two ranks of bank i; N: the number of banks

Worked Example: Banca Transilvania and the BET

- TLV and the BET, daily log returns since 2010 ($n = 3\,414$); quantile regression at 1%: $\hat{b} = 0.61$
  - TLV: $\hat{q}_{1\%} = -5.23\%$ (VaR 1% of TLV), median 0.18%
- ΔCoVaR 1% $= 0.61 \times (0.18 + 5.23) = 3.31\%$; CoVaR 1% of the BET $= 5.28\%$
  - 95% block-bootstrap interval for ΔCoVaR: [2.35, 4.78]
- Interpretation: a distress day of TLV moves the 1% quantile of the BET down by about 3.31 percentage points
  - part of this is mechanical: TLV is one of the largest weights in the BET; a cleaner system would be the BET without TLV

VaR Is Not ΔCoVaR



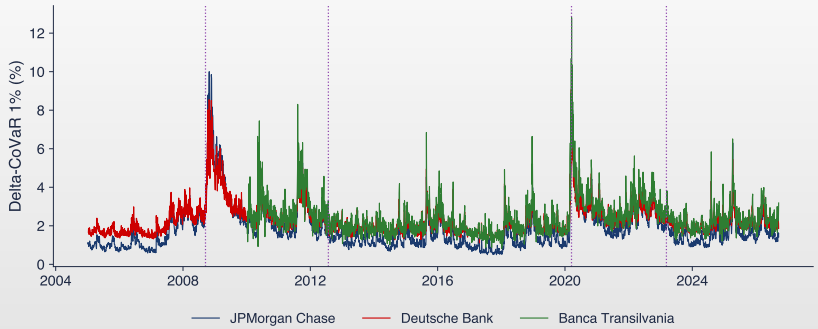
- Each point: a bank, its own VaR 1% (horizontal) and its ΔCoVaR 1% (vertical); Spearman rank correlation -0.31
- Interpretation: Citigroup and Bank of America are among the three largest VaR 1% and among the three smallest ΔCoVaR; regulating banks by their own VaR misses systemic risk Adrian & Brunnermeier (2016)

SFM_ch15_covar

Time-Varying ΔCoVaR

- Adrian and Brunnermeier let the quantiles depend on lagged **state variables** M_{t-1} Adrian & Brunnermeier (2016)
  - M_{t-1}: the vector of state variables known at the close of day $t-1$ (market stress, interest rates)
  - $X_t^i = c + \gamma^\top M_{t-1}$ at levels α and 50%: $\hat{q}_{\alpha,t}(X^i)$ and $\hat{q}_{50\%,t}(X^i)$
  - $X_t^{sys} = a + bX_t^i + \delta^\top M_{t-1}$ at level α; then $\Delta\mathrm{CoVaR}_t = \hat{b}\,(\hat{q}_{50\%,t} - \hat{q}_{\alpha,t})$
  - c, γ, a, b, δ: coefficients of the quantile regressions (δ is not the δ of Forbes–Rigobon)
- The paper uses seven state variables (VIX, liquidity and credit spreads, interest rates, market and real-estate returns)
  - here, a reduced set: the VIX level and the index return of the previous day
- The slope $\hat{b}$ is fixed; the time variation comes from the distance between the two conditional quantiles of the bank

ΔCoVaR 1% through Time



- JPMorgan Chase: average 1.86%, maximum 10.07% (18 March 2020); Deutsche Bank: maximum 9.08% (13 March 2020); Banca Transilvania: maximum 12.81% (17 March 2020)
- Interpretation: the contribution of a bank to systemic risk is not a constant; it jumps when the VIX jumps (2008, March 2020) and is small in calm years (JPM in 2017: 0.75%)

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Case Study: Adrian and Brunnermeier (2016)

- Adrian & Brunnermeier (2016): CoVaR of US financial institutions (commercial and investment banks, insurers, brokers), weekly data 1986–2010
  - the system: the portfolio of all financial institutions; seven state variables
- Findings
  - in the cross-section, the VaR of an institution says little about its ΔCoVaR
  - **forward** ΔCoVaR: size, leverage and maturity mismatch two years earlier predict the contribution in a crisis
  - a basis for countercyclical capital: charge banks when risk is built up, not when it shows



Tobias Adrian

Photo: Chianoo Adrian (2025); CC BY 4.0; Wikimedia Commons



Markus Brunnermeier

Photo: Iitkgpswapnil (2015); CC BY-SA 4.0; Wikimedia Commons

Marginal Expected Shortfall

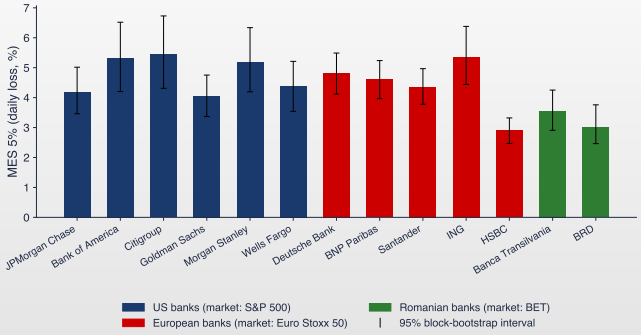
- **MES** Acharya, Pedersen, Philippon & Richardson (2017): the expected loss of institution i on the days when the market is in its tail
  - $\mathrm{MES}^i_\alpha = -E\left[X^i \mid X^m \le q_\alpha(X^m)\right]$; we use the worst 5% of market days: "MES 5%"
  - X^m: the market return; $q_\alpha(X^m)$: its α-quantile; the bar: "on the days when"
  - the direction is the opposite of CoVaR: the bank given the market, not the market given the bank
- Why "marginal": ES of the market $= \sum_i w_i \,\mathrm{MES}^i$ when the market is a portfolio with weights w_i
  - MES is the contribution of i to the ES of the system (Chapter 10)
- Estimation: sort the days by the market return, take the worst $\lceil n\alpha \rceil$ days, average the bank's returns on those days, change the sign

Worked Example: MES from Ten Days

Day	1	2	3	4	5	6	7	8	9	10
market	-4.0	1.2	-0.5	2.1	-6.0	0.3	-1.1	0.8	-2.2	1.5
bank	-6.5	1.8	-0.2	2.9	-9.0	0.1	-2.0	1.1	-3.1	2.4

- Level $\alpha = 20\%$: $\lceil 10 \times 0.2 \rceil = 2$ worst market days: day 5 (-6.0) and day 1 (-4.0)
  - MES 20% $= -(-9.0 - 6.5)/2 = 7.75\%$; ES 20% of the market $= -(-6.0 - 4.0)/2 = 5.00\%$
- Interpretation: on the market's bad days this bank loses about 1.5 times as much as the market: it amplifies the crashes
- Note: MES selects the days by the market return, not by the bank return

MES 5% of 13 Banks



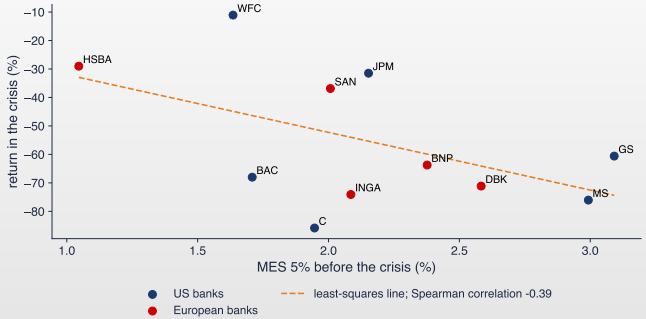
- Average daily loss of each bank on the worst 5% of days of its regional index (US and Europe since 2005, Romania since 2010); 95% block-bootstrap intervals
- Largest: Citigroup (5.44%) and ING (5.35%); smallest: HSBC (2.90%); TLV 3.54%, BRD 3.00%

SFM_ch15_mes_srisk

Interpretation: MES and ΔCoVaR

- MES is close to a tail beta: the average market loss on its worst 5% of days times the sensitivity of the bank
  - Citigroup: OLS beta 1.66, MES 5.44%; HSBC: beta 0.81, MES 2.90%
- MES and ΔCoVaR rank the 13 banks differently: Spearman correlation -0.29
  - they answer different questions: "who suffers in a crash" against "who adds to the crash"
  - a regulator should look at several measures, with their intervals Benoit, Colliard, Hurlin & Pérignon (2017)

Case Study: Did MES Predict the 2008 Losses?



- Design of Acharya, Pedersen, Philippon & Richardson (2017): MES 5% measured from June 2006 to June 2007, against the return from July 2007 to December 2008; here 11 US and European banks
- Spearman correlation -0.39 ($p = 0.23$, $n = 11$): the sign of the paper (higher MES, larger loss), but not significant with 11 banks; worst return: Citigroup (-86%)

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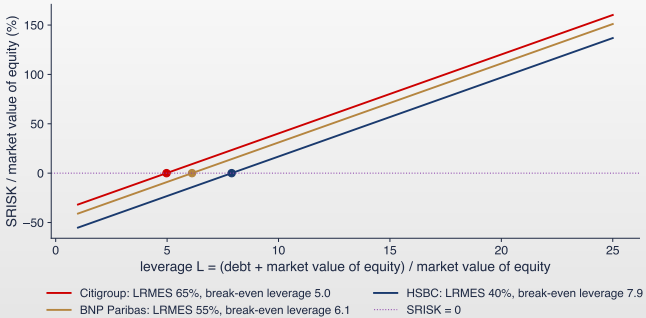
From MES to LRMES

- A crisis lasts months: we need the loss of the bank if the market falls 40% over six months: **LRMES** (long-run MES)
  - approximation of Acharya, Engle & Richardson (2012): $\mathrm{LRMES} \approx 1 - \exp(-18 \times \mathrm{MES})$, with MES the daily loss (as a fraction) on the days when the market falls by more than 2%
  - the factor 18 turns the daily loss on bad days into the loss over a six-month crisis; $1 - e^{-x}$ keeps LRMES between 0 and 1
- Example: MES $= 3\% = 0.03$: $\mathrm{LRMES} = 1 - e^{-0.54} = 1 - 0.583 = 41.7\%$
  - the bank would lose about 41.7% of its market value in such a crisis
- Our 13 banks: LRMES between 40% and 65%; Brownlees and Engle estimate it instead from a dynamic model (GARCH and dynamic correlation) Brownlees & Engle (2017)

SRISK: the Capital Shortfall in a Crisis

- A bank needs equity of at least a fraction k of its assets; $k = 8\%$ in Brownlees & Engle (2017)
  - D: book value of debt; W: market value of equity; assets $\approx D + W$
- **SRISK** $= k(D + W_{\text{crisis}}) - W_{\text{crisis}}$ with $W_{\text{crisis}} = (1 - \text{LRMES})W$:
  - $\text{SRISK} = kD - (1 - k)(1 - \text{LRMES})\, W$ Brownlees & Engle (2017)
  - kD: the capital required on the debt; $(1 - k)(1 - \text{LRMES})W$: the equity left after the crisis, net of the capital required on it
  - SRISK > 0: capital missing in a crisis; the **aggregate SRISK** (sum of the positive values) is what the state might have to cover
- Example: $D = 900$, $W = 100$ (billion), LRMES $= 41.7\%$
  - SRISK $= 0.08 \times 900 - 0.92 \times (1 - 41.7\%) \times 100 = 72 - 53.6 = 18.4$ billion

SRISK and Leverage



- SRISK$/W = k(L-1) - (1-k)(1-\mathrm{LRMES})$ with leverage $L = (D+W)/W$; SRISK > 0 above $L^* = 1 + (1-k)(1-\mathrm{LRMES})/k$
- Interpretation: Citigroup needs capital in a crisis above a leverage of 5.0, HSBC only above 7.9; banks typically have L between 10 and 20, so leverage decides the sign

SFM_ch15_mes_srisk

Case Study: SRISK and V-Lab

- Brownlees & Engle (2017): SRISK of large US financial firms, with LRMES from a GARCH–DCC model (Chapter 9)
  - aggregate SRISK rose before the failure of Lehman; it helps to forecast falls in industrial production and rises in unemployment
  - the ranking by SRISK in 2007–2008 points to the institutions that later needed public support
- Europe: Engle, Jondeau & Rockinger (2015); a lower $k = 5.5\%$ is used because of the accounting rules (IFRS, International Financial Reporting Standards)
  - published and updated on V-Lab SRISK (NYU Stern) for financial institutions from many countries



Viral Acharya

Photo: MeJudice (2014); CC BY 3.0; Wikimedia Commons



Robert Engle

Photo: Jon Demske (2022); CC BY-SA 4.0; Wikimedia Commons

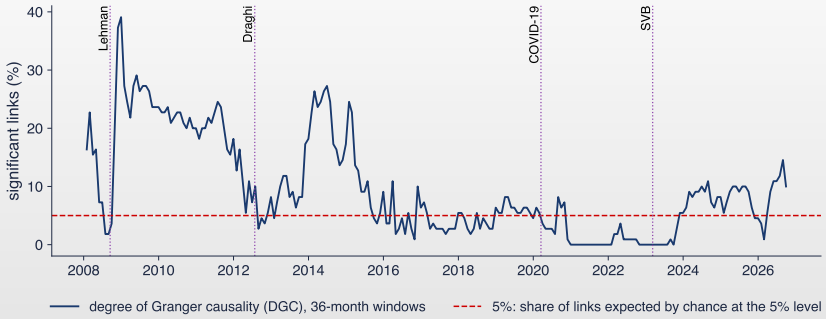
The Network View

- A **network**: nodes (banks) and directed links ("a shock to i affects j")
  - real links (interbank loans) are mostly confidential; we estimate links from returns
- Two statistical tools
  - **Granger causality**: do past returns of i help to forecast j? Billio, Getmansky, Lo & Pelizzon (2012)
  - **variance decompositions**: what share of the forecast error of j comes from shocks to i? Diebold & Yilmaz (2014)
- A statistical link is a co-movement, not a contract: it shows where to look, not why

Granger Causality

- x **Granger-causes** y if past values of x improve the forecast of y beyond the past of y Granger (1969)
  - regression $y_t = a + b\,y_{t-1} + c\,x_{t-1} + e_t$; test $H_0 : c = 0$ with a t test at 5%
  - a: the constant; b: the own past of y; c: the effect of yesterday's x; e_t: the error; $c \neq 0$: a link from x to y
- **Degree of Granger causality** (DGC) Billio, Getmansky, Lo & Pelizzon (2012): the share of the $N(N-1)$ ordered pairs with a significant link
  - N: the number of banks; $N(N-1)$: the ordered pairs ($i \to j$ and $j \to i$ count separately)
  - the paper: monthly returns, 36-month rolling windows, lag 1, 5% level
  - the paper also filters each series with a GARCH(1,1) model; with 36 monthly observations per window we report the unfiltered test
- Under no links, about 5% of pairs are significant by chance: DGC must be compared with 5%

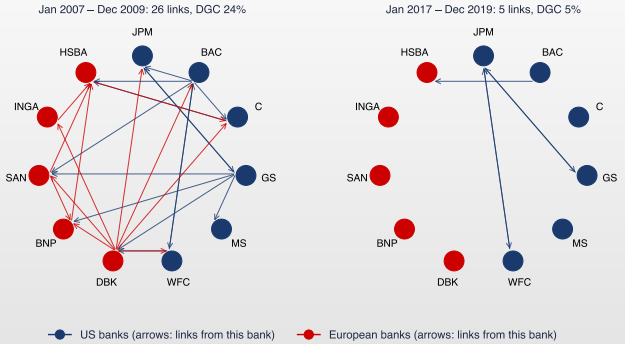
Granger Networks through Time



- 11 US and European banks, monthly returns since 2005 (260 months); first window ends in January 2008
- DGC: maximum 39% (window ending 31 December 2008), average 10%, minimum 0% (31 December 2020)

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Two Networks: 2007–2009 and 2017–2019



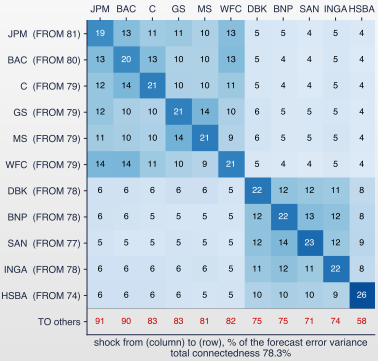
- Arrows: significant Granger-causal links at 5% in a 36-month window; 26 links in the crisis window, 5 in the calm one
- Interpretation: as in Billio, Getmansky, Lo & Pelizzon (2012), the banks became more connected around the crisis; in calm years the network is close to what chance alone would give

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Diebold–Yilmaz Connectedness

- Fit a VAR (vector autoregression) to the returns of N banks; forecast H days ahead
  - VAR(p): $x_t = c + \Phi_1 x_{t-1} + \cdots + \Phi_p x_{t-p} + u_t$; x_t: the vector of the N returns; Φ_j: $N \times N$ coefficient matrices; u_t: the shocks
  - d_{ij}: the share of the H-day forecast error variance of bank i due to shocks to bank j (generalized decomposition, Pesaran & Shin (1998))
  - each row sums to 100%
- Measures Diebold & Yilmaz (2012); Diebold & Yilmaz (2014)
  - FROM others: $\sum_{j \neq i} d_{ij}$; TO others: $\sum_{j \neq i} d_{ji}$; NET = TO − FROM
  - **total connectedness**: $\frac{1}{N} \sum_{i \neq j} d_{ij}$, the average share of risk that comes from the others
- Here: daily returns of 11 banks since 2005, VAR order by BIC (Bayesian information criterion), $H = 10$ days

The Connectedness Table



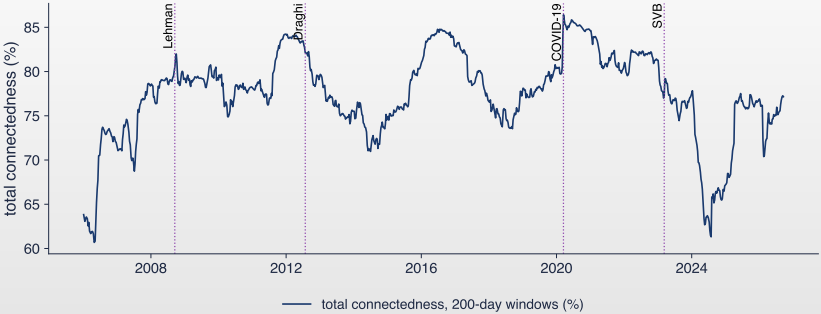
- 4 895 common days, VAR(1); total connectedness 78.3%; two blocks: US banks are connected mostly with US banks, European with European
- Largest transmitter (TO): JPMorgan Chase; smallest: HSBC (NET −15.8)

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Interpretation of the Connectedness Table

- Each bank receives about four fifths of its forecast error variance from the others (FROM between 74 and 81)
  - daily bank returns are dominated by common shocks; the own share (the diagonal) is only about one fifth
- NET: US banks are net transmitters (JPMorgan Chase $+10.6$), European banks net receivers (HSBC -15.8)
  - partly timing: New York closes after Europe, so US news reaches European prices the next day
- Connectedness describes the data; it is not a map of exposures between the banks

Total Connectedness through Time



- Rolling 200-day windows moved by 5 days; minimum 61% (28 April 2006), maximum 86% (17 March 2020); October 2008 – March 2009: 79%
- Interpretation: connectedness rose during 2007, before Lehman, and reaches its maximum in March 2020; but in 2016, a year without a crisis in the data, it reaches 85%, above the 2008–2009 average: a high level alone does not signal a crisis; it falls when bank-specific news dominate (2006, 2024)

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Further Reading: the Financial Risk Meter

- **FRM** (Financial Risk Meter) Mihoci, Althof, Chen & Härdle (2020): a daily index of systemic risk from quantile regressions with many regressors
  - for each bank: a lasso quantile regression (Chapter 13) of its returns on the returns of all other banks and on macro variables
  - the penalty λ needed to fit the tail is large when the banks move together: FRM = the average λ
- Extensions
  - TENET (tail-event driven network) Härdle, Wang & Yu (2016): CoVaR with many banks, a network of tail links
  - FRM for emerging markets Ben Amor, Althof & Härdle (2022); the video course on Quantinar (Reading and Tools)

Micro- and Macroprudential

- **Microprudential**: is each bank safe on its own? (capital, VaR, backtesting, Chapter 10)
- **Macroprudential**: is the system safe? Two dimensions
  - **in time**: risk builds up in booms (credit growth, leverage) and shows in busts: build buffers in good times
  - **across institutions**: larger and more connected banks need more capital
- The measures of this chapter (CoVaR, SRISK, connectedness) are among the tools that central banks use to monitor the second dimension

Basel III Capital Buffers

- Minimum CET1 (common equity tier 1) capital: 4.5% of risk-weighted assets BCBS (2011); on top of it, buffers Basel Framework, RBC30:
  - **capital conservation buffer**: 2.5%; below it, dividends and bonuses are restricted
  - **countercyclical buffer**: 0–2.5%, raised when credit grows too fast, released in a crisis
- For systemic importance
  - **G-SIB surcharge**: 1% to 3.5%, by a score of size, interconnectedness, substitutability, complexity and cross-border activity BCBS (2018)
  - in the EU: O-SII (other systemically important institutions) buffers and a systemic risk buffer



The BIS tower in Basel

Photo: Taxiarchos228 (2014); CC BY-SA 4.0; Wikimedia Commons

Who Does It: the ESRB and the BNR

- **ESRB** (European Systemic Risk Board), set up in 2010 after the crisis, chaired by the President of the ECB (ESRB website)
  - monitors systemic risk in the EU, issues warnings and recommendations to national authorities
- Romania: **CNSM** (Comitetul Național pentru Supravegherea Macroprudențială), with the BNR, the ASF (financial supervisory authority) and the Government (CNSM website)
  - the BNR sets the capital buffers of the banks (countercyclical, O-SII, systemic risk buffer) on CNSM recommendations
  - the BNR also uses limits on borrowers (loan-to-value, debt service-to-income) and publishes a financial stability report



The BNR palace in Bucharest

Photo: Britchi Mirela (2018); CC BY-SA 4.0; Wikimedia Commons

An Open Question

- **Is the systemic risk of the Romanian banks imported from their euro-area parents, or is it local?**
  - today: ΔCoVaR of TLV peaks in March 2020 (12.81%); BRD belongs to a French group, TLV is Romanian-owned
  - the BET is partly the banks themselves, so a better "system" is needed
- Why it is open: two banks, few crises, different trading hours (Bucharest, Frankfurt, New York), thin trading in some years
- AI tools can speed up such a study; they do not replace checking it Wang et al. (2023)

How AI Could Help

- **Literature**: a list of studies on systemic risk in Central and Eastern Europe and on parent–subsidiary contagion
- **Code**: a first draft of ΔCoVaR of TLV and BRD with the euro-area bank portfolio as the conditioning variable and state variables
- **Robustness**: two-day returns (trading hours), weekly data, CoVaR at 5%, other state variables
- Example prompt
  - `Write Python code that estimates the Delta-CoVaR at 1% of the BET conditional on Banca Transilvania by quantile regression, with the lagged VIX and the lagged Euro Stoxx 50 return as state variables, and adds a moving-block bootstrap interval.`

What to Check

- The level convention: "CoVaR 1%" and "MES 5%" name the tail probability; CoVaR and MES are positive losses
- The direction of conditioning: CoVaR is the system given the bank, MES is the bank given the market
- ΔCoVaR compares distress with the **median** of the bank, not with the bank's VaR
- Prices joined on common days before returns; BVB days without trades removed; different trading hours
- Uncertainty: quantile regressions at 1% rest on about 1% of the days; report bootstrap intervals
- References: every cited paper must exist; check the DOI

Project Idea

- **Question**: do the euro-area banks raise the tail risk of the Romanian banks in stress periods?
  - data: TLV, BRD, BET, the five European banks, Euro Stoxx 50, VIX, since 2010 (EODHD)
- Steps
  - ΔCoVaR 1% of each Romanian bank conditional on the European bank portfolio, static and with state variables
  - MES 5% of TLV and BRD with the Euro Stoxx 50 as the market; Diebold–Yilmaz FROM shares on rolling windows
  - compare 2011–2012, March 2020 and March 2023, with block-bootstrap intervals
- Deliverable: one table, one chart, and a paragraph on what the data can and cannot show
- Declare any AI use, and list the errors of the AI that you corrected

Key Takeaways

- Systemic risk is a property of the system: contagion, common exposures, fire sales and runs, amplified by leverage
- In crises bank correlations and tail dependence rise; correct for volatility before speaking of contagion
- CoVaR 1%: the system given the bank; $\Delta\text{CoVaR} = \hat{b}(\hat{q}_{50\%} - \hat{q}_{\alpha})$ from quantile regression
- MES 5%: the bank given the market; $\text{SRISK} = kD - (1-k)(1-\text{LRMES})W$, driven by leverage
- Networks (Granger, Diebold–Yilmaz) show more connectedness around crises
- Macroprudential policy: Basel III buffers, G-SIB and O-SII surcharges; the ESRB in the EU, the CNSM and the BNR in Romania

Key Formulas

Quantity	Formula
CoVaR	$\mathrm{CoVaR}_\alpha = -(\hat{a} + \hat{b}\,\hat{q}_\alpha(X^i))$, $X^{sys} = a + bX^i$ at level α
ΔCoVaR	$\hat{b}\,(\hat{q}_{50\%}(X^i) - \hat{q}_\alpha(X^i))$
MES	$-E[X^i \mid X^m \le q_\alpha(X^m)]$
LRMES	$1 - \exp(-18\,\mathrm{MES}_{2\%})$
SRISK	$kD - (1-k)(1-\mathrm{LRMES})W$, $L^* = 1 + (1-k)(1-\mathrm{LRMES})/k$
Forbes–Rigobon	$\rho^* = \rho/\sqrt{1+\delta(1-\rho^2)}$
Granger	$y_t = a + by_{t-1} + cx_{t-1} + e_t$, $H_0 : c = 0$
Diebold–Yilmaz	total $= \frac{1}{N}\sum_{i\neq j} d_{ij}$, $\mathrm{NET}_i = \mathrm{TO}_i - \mathrm{FROM}_i$

Check Yourself

- **Question**: bank A has a larger VaR 1% than bank B. Is A more systemic?
  - **Answer**: not necessarily: systemic importance depends on the link with the system (ΔCoVaR, MES), not on the bank's own tail
- **Question**: a quantile regression at 1% gives $\hat{b} = 0.5$, $\hat{q}_{1\%}(X^i) = -6\%$, $\hat{q}_{50\%}(X^i) = 0$. What is ΔCoVaR?
  - **Answer**: $0.5 \times (0 - (-6)) = 3\%$
- **Question**: why can SRISK be negative?
  - **Answer**: a bank with low leverage keeps enough capital even after a market fall of 40%: it has a capital surplus
- Next: Chapter 16, review

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